

VAJRA Security Platform - Complete Data Flow Diagram

This document shows the complete data flow from user login through all system modules to logout.

VAJRA Security Platform - Complete Data Flow Diagram

User Journey from Login to Logout

graph TD

Start([User Opens Application]) --> LoginScreen[Login Screen]

LoginScreen --> UserInput{User Enters Credentials}

UserInput -->|Valid Credentials| AuthAPI[POST /api/auth/login]

UserInput -->|Invalid Credentials| LoginError[Show Error Message]

LoginError --> LoginScreen

AuthAPI --> JWTGeneration[Generate JWT Tokens]

JWTGeneration --> TokenStorage[Store Tokens in LocalStorage]

TokenStorage --> RedirectDashboard[Redirect to Dashboard]

RedirectDashboard --> DashboardLoad[Dashboard Component Loads]

DashboardLoad --> AuthCheck{Check Authentication}

AuthCheck -->|Not Authenticated| LoginScreen

AuthCheck -->|Authenticated| DashboardInit[Initialize Dashboard]

DashboardInit --> ParallelLoad[Load Dashboard Data]

ParallelLoad --> SummaryAPI[GET /api/dashboard/summary]

ParallelLoad --> AlertsAPI[GET /api/dashboard/alerts]

ParallelLoad --> AttackMapAPI[GET /api/dashboard/attack-map]

SummaryAPI --> AlienVaultService[AlienVault Service]

AlienVaultService --> AlienVaultAPI[AlienVault OTX API]

AlienVaultAPI --> SummaryData[Threat Intelligence Data]

SummaryData --> DisplaySummary[Display Summary Cards]

AlertsAPI --> MockAlerts[Mock Alert Data]

MockAlerts --> DisplayAlerts[Display Critical Alerts]

AttackMapAPI --> MockMap[Mock Map Data]

MockMap --> DisplayMap[Display Attack Map]

DisplaySummary --> DashboardReady[Dashboard Ready]

DisplayAlerts --> DashboardReady

DisplayMap --> DashboardReady

DashboardReady --> UserNavigation{User Navigation}

UserNavigation -->|Threat Intelligence| ThreatModule[Threat Intelligence Module]

UserNavigation -->|Ransomware| RansomwareModule[Ransomware Module]

UserNavigation -->|News| NewsModule[News Module]

UserNavigation -->|Companies| CompaniesModule[Companies Module]

UserNavigation -->|Domain Analysis| DomainModule[Domain Analysis Module]

UserNavigation -->|Reports| ReportsModule[Reports Module]

UserNavigation -->|Logout| LogoutFlow[Logout Flow]

%% Threat Intelligence Module Flow

ThreatModule --> ThreatAPI[GET /api/threat-intelligence]

ThreatModule --> ThreatTrendAPI[GET /api/threat-intelligence/trend]

ThreatModule --> ThreatActorsAPI[GET /api/threat-intelligence/actors]

ThreatModule --> ThreatIndustriesAPI[GET /api/threat-intelligence/industries]

ThreatAPI --> AlienVaultThreat[AlienVault Threat Data]

ThreatTrendAPI --> MockTrend[Mock Trend Data]

ThreatActorsAPI --> ThreatActorsDB[Database Query]

ThreatIndustriesAPI --> IndustriesDB[Database Query]

AlienVaultThreat --> DisplayThreat[Display Threat Data]

MockTrend --> DisplayTrend[Display Trend Charts]

ThreatActorsDB --> DisplayActors[Display Threat Actors]

IndustriesDB --> DisplayIndustries[Display Industry Data]

DisplayThreat --> ThreatComplete[Threat Module Complete]

DisplayTrend --> ThreatComplete

DisplayActors --> ThreatComplete

DisplayIndustries --> ThreatComplete

ThreatComplete --> UserNavigation

%% Ransomware Module Flow

RansomwareModule --> RansomwareAPI[GET /api/ransomware]

RansomwareModule --> RansomwareStatsAPI[GET /api/ransomware/stats]

RansomwareModule --> RansomwareGroupAPI[GET /api/ransomware/group/{name}]

RansomwareAPI --> RansomwareService[Ransomware Service]

RansomwareStatsAPI --> RansomwareStatsService[Ransomware Stats Service]

RansomwareGroupAPI --> RansomwareGroupService[Ransomware Group Service]

RansomwareService --> RansomwareLiveAPI[Ransomware.live API]

RansomwareStatsService --> RansomwareLiveStatsAPI[Ransomware.live Stats API]

RansomwareGroupService --> RansomwareLiveGroupAPI[Ransomware.live Group API]

RansomwareLiveAPI --> RansomwareData[Ransomware Incident Data]

RansomwareLiveStatsAPI --> RansomwareStatsData[Ransomware Statistics]

RansomwareLiveGroupAPI --> RansomwareGroupData[Ransomware Group Data]

RansomwareData --> DisplayRansomware[Display Ransomware Incidents]

RansomwareStatsData --> DisplayRansomwareStats[Display Ransomware Stats]

RansomwareGroupData --> DisplayRansomwareGroup[Display Group Data]

DisplayRansomware --> RansomwareComplete[Ransomware Module Complete]

DisplayRansomwareStats --> RansomwareComplete

DisplayRansomwareGroup --> RansomwareComplete

RansomwareComplete --> UserNavigation

%% News Module Flow

NewsModule --> NewsAPI[GET /api/news]

NewsAPI --> HackerNewsService[Hacker News Service]

HackerNewsService --> HackerNewsAPI[Hacker News API]

HackerNewsAPI --> NewsData[News Stories Data]

NewsData --> FilterNews[Filter Cybersecurity Keywords]

FilterNews --> DisplayNews[Display News Feed]

DisplayNews --> NewsComplete[News Module Complete]

NewsComplete --> UserNavigation

%% Companies Module Flow

CompaniesModule --> CompaniesListAPI[GET /api/companies]

CompaniesModule --> CompanyDetailAPI[GET /api/companies/{id}]

CompaniesModule --> AddCompanyAPI[POST /api/companies]

CompaniesListAPI --> CompaniesDB[Database Query]

CompanyDetailAPI --> CompanyDetailDB[Database Query with Risk Assessment]

AddCompanyAPI --> CompanyCreate[Create Company in Database]

CompaniesDB --> DisplayCompanies[Display Company List]

CompanyDetailDB --> DisplayCompanyDetail[Display Company Details]

CompanyCreate --> CompanyCreated[Company Created Successfully]

DisplayCompanies --> CompaniesComplete[Companies Module Complete]

DisplayCompanyDetail --> CompaniesComplete

CompanyCreated --> CompaniesComplete

CompaniesComplete --> UserNavigation

%% Domain Analysis Module Flow

DomainModule --> DomainRiskAPI[GET /api/domain-risk/{domain}]

DomainModule --> DomainAnalysisAPI[GET /api/domain-analysis/{domain}]

DomainRiskAPI --> DomainRiskService[Domain Risk Service]

DomainAnalysisAPI --> DomainAnalysisService[Domain Analysis Service]

DomainRiskService --> ExternalDomainAPI[External Domain APIs]

DomainAnalysisService --> ExternalDomainAnalysisAPI[External Domain Analysis APIs]

ExternalDomainAPI --> DomainRiskData[Domain Risk Data]

ExternalDomainAnalysisAPI --> DomainAnalysisData[Domain Analysis Data]

DomainRiskData --> DisplayDomainRisk[Display Domain Risk]

DomainAnalysisData --> DisplayDomainAnalysis[Display Domain Analysis]

DisplayDomainRisk --> DomainComplete[Domain Module Complete]

DisplayDomainAnalysis --> DomainComplete

DomainComplete --> UserNavigation

%% Reports Module Flow

ReportsModule --> ReportSelection{Select Report Type}

ReportSelection -->|Threat Intelligence| ThreatReportAPI[GET /api/reports/threat-intelligence]

ReportSelection -->|Ransomware| RansomwareReportAPI[GET /api/reports/ransomware]

ReportSelection -->|Global Attacks| GlobalAttacksReportAPI[GET /api/reports/global-attacks]

ReportSelection -->|Company| CompanyReportAPI[GET /api/reports/company/{id}]

ReportSelection -->|Executive| ExecutiveReportAPI[GET /api/reports/executive]

ReportSelection -->|Comprehensive| ComprehensiveReportAPI[GET /api/reports/comprehensive]

ThreatReportAPI --> PDFServiceThreat[PDF Service - Threat Intelligence]

RansomwareReportAPI --> PDFServiceRansomware[PDF Service - Ransomware]

GlobalAttacksReportAPI --> PDFServiceGlobal[PDF Service - Global Attacks]

CompanyReportAPI --> PDFServiceCompany[PDF Service - Company]

ExecutiveReportAPI --> PDFServiceExecutive[PDF Service - Executive]

ComprehensiveReportAPI --> PDFServiceComprehensive[PDF Service - Comprehensive]

PDFServiceThreat --> PDFGenerationThreat[Generate PDF with Black/White Styling]

PDFServiceRansomware --> PDFGenerationRansomware[Generate PDF with Black/White Styling]

PDFServiceGlobal --> PDFGenerationGlobal[Generate PDF with Black/White Styling]

PDFServiceCompany --> PDFGenerationCompany[Generate PDF with Black/White Styling]

PDFServiceExecutive --> PDFGenerationExecutive[Generate PDF with Black/White Styling]

PDFServiceComprehensive --> PDFGenerationComprehensive[Generate Combined PDF]

PDFGenerationThreat --> DownloadThreat[Download Threat Intelligence PDF]

PDFGenerationRansomware --> DownloadRansomware[Download Ransomware PDF]

PDFGenerationGlobal --> DownloadGlobal[Download Global Attacks PDF]

PDFGenerationCompany --> DownloadCompany[Download Company PDF]

PDFGenerationExecutive --> DownloadExecutive[Download Executive PDF]

PDFGenerationComprehensive --> DownloadComprehensive[Download Comprehensive PDF]

DownloadThreat --> ReportsComplete[Reports Module Complete]

DownloadRansomware --> ReportsComplete

DownloadGlobal --> ReportsComplete

DownloadCompany --> ReportsComplete

DownloadExecutive --> ReportsComplete

DownloadComprehensive --> ReportsComplete

ReportsComplete --> UserNavigation

%% Logout Flow

LogoutFlow --> LogoutAPI[POST /api/auth/logout]

LogoutAPI --> ClearTokens[Clear Tokens from LocalStorage]

ClearTokens --> RedirectLogin[Redirect to Login Screen]

RedirectLogin --> End([Session Ended])

%% WebSocket Real-time Updates

DashboardReady --> WebSocketConnect[Connect to WebSocket /ws]

WebSocketConnect --> WebSocketEvents{WebSocket Events}

WebSocketEvents -->|Threat Updates| ThreatUpdate[Update Threat Data]

WebSocketEvents -->|Alert Notifications| AlertNotification[Show New Alert]

WebSocketEvents -->|Dashboard Refresh| DashboardRefresh[Refresh Dashboard]

ThreatUpdate --> UpdateThreatDisplay[Update Threat Display]

AlertNotification --> UpdateAlertDisplay[Update Alert Display]

DashboardRefresh --> RefreshDashboardData[Refresh All Dashboard Data]

UpdateThreatDisplay --> DashboardReady

UpdateAlertDisplay --> DashboardReady

RefreshDashboardData --> DashboardReady

%% Background Data Sync

DashboardReady --> BackgroundSync[Background Data Sync]

BackgroundSync --> Scheduler[Scheduler Service]

Scheduler --> PeriodicUpdates[Periodic Data Updates]

PeriodicUpdates --> UpdateAlienVault[Update AlienVault Data]

PeriodicUpdates --> UpdateRansomware[Update Ransomware Data]

PeriodicUpdates --> UpdateNews[Update Hacker News Data]

UpdateAlienVault --> CacheAlienVault[Cache Updated Data]

UpdateRansomware --> CacheRansomware[Cache Updated Data]

UpdateNews --> CacheNews[Cache Updated Data]

CacheAlienVault --> WebSocketEvents

CacheRansomware --> WebSocketEvents

CacheNews --> WebSocketEvents

style Start fill:#4CAF50,color:#fff

style End fill:#f44336,color:#fff

style LoginScreen fill:#2196F3,color:#fff

style DashboardReady fill:#FF9800,color:#fff

style UserNavigation fill:#9C27B0,color:#fff

style ThreatComplete fill:#4CAF50,color:#fff

style RansomwareComplete fill:#4CAF50,color:#fff

style NewsComplete fill:#4CAF50,color:#fff

style CompaniesComplete fill:#4CAF50,color:#fff

style DomainComplete fill:#4CAF50,color:#fff

style ReportsComplete fill:#4CAF50,color:#fff

Detailed Component Data Flow

Authentication Flow

User → Login Form → Frontend Validation → API Call → Backend Validation
→ Database Query → JWT Generation → Token Storage → Dashboard Redirect

Dashboard Data Flow

Dashboard Component → Multiple API Calls → External Services (AlienVault)
→ Data Processing → State Management → Component Rendering → Real-time Updates

Threat Intelligence Flow

Threat Module → API Requests → AlienVault Service → External API
→ Data Transformation → State Update → Visualization Display

Ransomware Tracking Flow

Ransomware Module → API Requests → Ransomware Service → Ransomware.live API
→ Data Processing → Statistics Calculation → Incident Display

News Feed Flow

News Module → API Request → Hacker News Service → Hacker News API
→ Keyword Filtering → Relevance Scoring → News Display

Company Risk Assessment Flow

Companies Module → Database Query → Risk Assessment Calculation
→ Threat Analysis → Report Generation → Dashboard Display

Report Generation Flow

Report Selection → API Request → PDF Service → Data Aggregation
→ PDF Generation (Black/White Styling) → File Download

Real-time Updates Flow

WebSocket Connection → Event Listening → Data Updates → State Management
→ UI Re-render → User Notification

Background Sync Flow

Scheduler → Periodic Triggers → External API Calls → Data Caching
→ WebSocket Events → Client Updates

Database Operations Flow

User Operations:

- Register: Create User Record
- Login: Query User & Validate Password
- Update: Modify User Profile
- Delete: Remove User Record

Company Operations:

- Create: Add Company to Monitoring
- Read: Fetch Company Details & Risk Assessment
- Update: Modify Company Information
- Delete: Remove Company from Monitoring

Threat Operations:

- Create: Log New Threat
- Read: Query Threat Data
- Update: Update Threat Status
- Delete: Archive Old Threats

Alert Operations:

- Create: Generate Security Alerts
- Read: Fetch User Alerts
- Update: Mark as Read/Resolved
- Delete: Remove Old Alerts

External API Integration Flow

AlienVault OTX:

Frontend → Backend → AlienVault Service → OTX API → Threat Data → Processing → Display

Ransomware.live:

Frontend → Backend → Ransomware Service → Ransomware.live API → Incident Data → Processing → Display

Hacker News:

Frontend → Backend → Hacker News Service → HN API → News Data → Filtering → Display

State Management Flow

User Actions → Component State → Global State (Zustand) → API Calls
→ Response Processing → State Update → Component Re-render → UI Update

Error Handling Flow

API Error → Error Boundary → Error Logging → User Notification
→ Fallback Data → Graceful Degradation → Retry Mechanism

Security Flow

Request → JWT Validation → Permission Check → Rate Limiting
→ Input Validation → Processing → Response → Security Headers